so that we can input the description (in terms of the same independent variables we had in the dataset) of a car we spotted at the dealer's, and see if the price they are asking is fair (and hopefully avoid being tricked into paying too much for a wreck).

We need to choose the model we think could fit the data best[15]; to keep it simple, let's restrict to functions with a single parameter (it could be, f.i., the engine power), as shown in figure 16.18: we can choose a constant function (`y = m`, a line parallel to the x axis), a generic line of the form `y = mx + b`, a quadratic curve ($m_1 * x^2 + m_2 * x + b$) or even more complicated models.

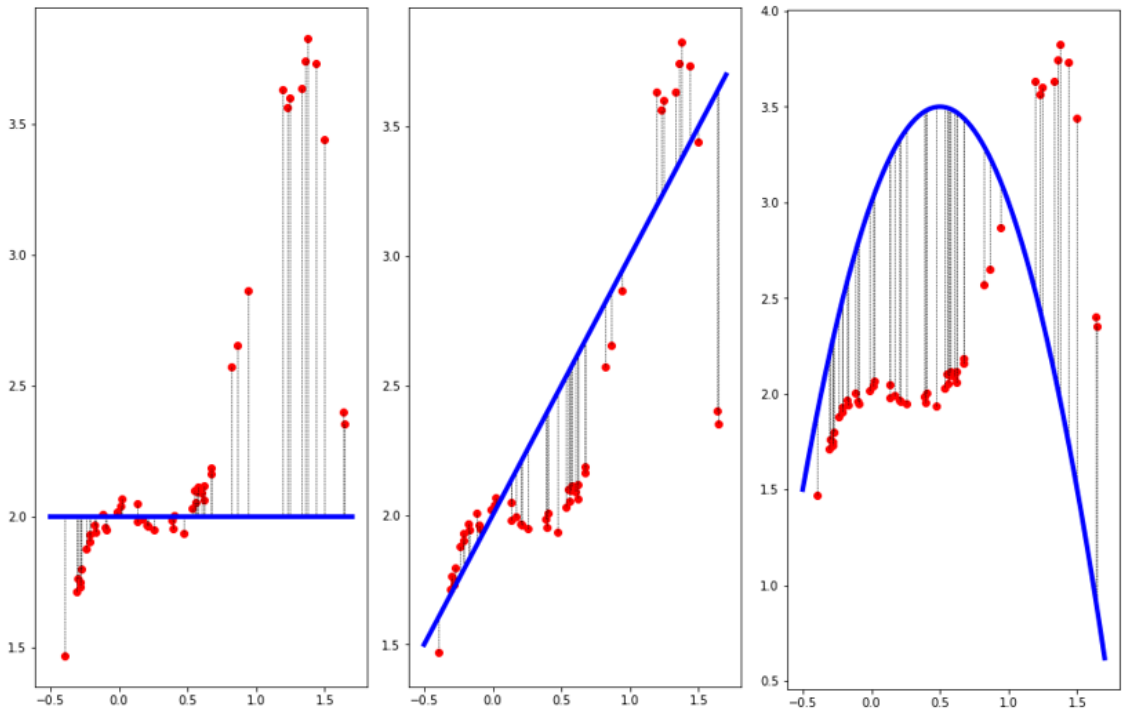


Figure 16.18 Linear regression on a dataset (points shown as red circles) using increasingly complex models: a constant, a line, and a quadratic curve. You can see that a higher-order model doesn't necessarily fit the data better.

The simpler the model, in general, the fewest data points are needed to "learn it": because after we choose the complexity of